```
public class TSP {
   public static void main(String[] args) {
      int[][] d = {
         {0,10,15,20},
         {10,0,35,25},
         {15,35,0,30},
         {20,25,30,0}
      };
      int[] path = {0,1,2,3};
      int cost = 0;
      System.out.println("Initial State:");
      System.out.println("A -> B -> C -> D");
      for (int i = 0; i < 4; i++)
         cost += d[path[i]][path[(i+1)%4]];
      System.out.println("\nGoal State:");
      System.out.println("A -> B -> C -> D -> A");
      System.out.println("Total Cost: " + cost);
   }
}
```

1. Theory

Travelling Salesman Problem (TSP):

TSP is a problem in which a salesman visits every city exactly once and returns to the starting city with minimum total distance.

Local Beam Search:

Local Beam Search is a search technique that maintains multiple candidate solutions instead of only one.

- It generates neighbouring solutions.

- Calculates the cost of each solution.
- Selects the best k solutions.
- Repeats until the best solution is found or the stopping condition is reached.

Important terms:

1. State: A complete tour of all cities.
2. Beam Width (k): Number of candidate solutions maintained.
3. Successor: A new tour generated from an existing tour.
4. Cost: Total distance of the tour.
5. Goal: Find a tour with minimum distance.

2. Algorithm

1. Start.
2. Read the distance matrix, beam width k and maximum iterations.
3. Generate k initial tours.
4. Calculate the cost of each tour.
5. Store the best tour.
6. Generate neighbouring tours.
7. Remove duplicate tours.
8. Sort tours according to their cost.
9. Select the best k tours.
10. Update the global best tour.
11. Repeat until the stopping condition is reached.
12. Display the best tour and total distance.
13. Stop.

3. Viva Questions and Answers

Q1. What is TSP?
TSP is a problem of visiting every city once and returning to the starting city with minimum distance.

Q2. Why is TSP called an optimization problem?
Because we need to find the minimum-cost tour.

Q3. What is local search?
It is a technique that improves a solution by searching its neighbouring solutions.

Q4. What is Local Beam Search?
It maintains multiple candidate solutions and selects the best ones.

Q5. What is the difference between Hill Climbing and Local Beam Search?
Hill Climbing maintains one solution, while Local Beam Search maintains multiple solutions.

Q6. What is beam width (k)?
It is the number of candidate solutions maintained at each iteration.

Q7. Why are duplicate tours removed?
To avoid checking the same solution again.

Q8. What is a 2-opt move?
It improves a tour by reversing a part of the route.

Q9. Why do we return to the starting city?
To complete the tour.

Q10. What is tour cost?
The total distance travelled through all cities, including the return journey.

Q11. What is a global best tour?
The best solution found during the entire search.

Q12. What is a successor?
A new solution generated from the current solution.

Q13. What is a random restart?
Starting the search again with a different initial solution.

Q14. What is the goal of TSP?
To find the shortest possible tour.